

Group Contribution Statement

Group E: Vishal Kumar, Avanti Gupta, Shivangi Manel

2026-05-03

MATH 167R Spring 2026 — Group E Contribution Statement

Vishal Kumar Responsible for data loading, data cleaning, and preprocessing steps including handling missing values, removing zero variance and highly correlated features, and applying min-max normalization. Also responsible for setting up the stratified sampling and train/test split.

Avanti Gupta Responsible for exploratory data analysis and visualizations including the class distribution plot, correlation heatmap, flow duration density plot, and Flow Bytes/s boxplot. Also responsible for computing and interpreting descriptive statistics.

Shivangi Manel Responsible for model training and evaluation including Random Forest and XGBoost implementation, feature importance analysis, F1 score comparison, and ROC curve analysis. Also responsible for writing the final report and compiling all results.